U.PORTO FACULDADE DE CIÊNCIAS UNIVERSIDADE DO PORTO	**Bachelor Degree in Bioinformatics** **Algorithms for the Analysis of Biological Sequences**	**2025/2026** **2nd Year** **2nd Semester**
INSTRUCTOR: Miriam Seoane Santos		

Research Workshop Guidelines

Objective

In this assignment, each group will **design and conduct an in-person research workshop** related to a topic explored in the course. Instead of a traditional presentation, your group will guide the class through a bioinformatics research problem and a practical activity using real biological data.

Your workshop should simulate a **mini research seminar with a hands-on component**, similar to tutorials often presented at scientific conferences.

This assignment encourages you to:

- Explore a bioinformatics topic beyond the material covered in the classes.
- Read and interpret scientific literature in bioinformatics.
- Investigate modern tools, algorithms, and analysis workflows used in bioinformatics.
- Communicate complex computational and biological concepts clearly.
- Train your skills in reproducible research ☺

The deliverables of this assignment are a **presentation (up to 10 slides) and your reproducible experiments**.

Instructions

- Groups should be composed of **4 students**.
- Each group should submit their deliverables on Moodle at most **one week after** their workshop.
- The presentations will occur during **PL classes**. The following groups will be available:
  - **Group A** - 16 April - Sequence Metrics and Similarity
  - **Group B** - 23 April - Pairwise Sequence Comparison
  - **Group C** - 30 April - Multiple Sequence Alignment
  - **Group D** - 14 May - Database Search Methods
  - **Group E** - 14 May - Molecular Evolution and Phylogeny
  - **Group F** - 21 May - Hidden Markov Models
  - **Group G** - 21 May - Motif Finding
- **The registration slots will open on the PL class (12 March).**

Workshop Structure

Each group will have about **45 min** to present their work (at most). You should prepare your session to include the following:

- **Context and Motivation (*approx. 10 min*):** introduce the biological research question/problem and explain why it is important and how bioinformatics helps answer the question. This can be based on a real-world bioinformatics research study.
- **Method, Strategy/Approach or Tool Overview (*approx. 15 min*):** explain the computational method or bioinformatics tool used in your activity.
- **Hands-on Activity (*approx. 20 min*):** prepare clear instructions and datasets to guide the class through a practical exercise using real biological data. Your goal should be providing a step-by-step experience, commenting on results obtained and providing explanations on your methodology.

Reproducibility

You should structure your pipeline in a way that it can be reproduced by others. You can deliver your data and code in the following ways:

- **.zip file on Moodle:** During the project submission, include a .zip file with your data, a Jupyter Notebook (or .py project), Snakemake files (or others), containing the complete code, and a .txt file with instructions on how to reproduce your results.

OR

- **GitHub repository:** Alternatively, you can push your code, data, and documentation to a public GitHub repository. In such case, please include a url in your presentation slides. Also, include a short README on your repository explaining how to reproduce your experiments.

Deliverables

The deliverables of this assignment are the **presentation slides** used during the workshop, along with your **reproducible experiments (datasets, scripts/notebooks, and instructions)**.

Evaluation Criteria

Your workshop will be evaluated based on the following criteria:

- **Creativity and Problem Definition (20%)**: Ability to integrate the course topics and define a clear and well-motivated question for investigation based of previous research.
- **Choice and Explanation of Computational Approach (25%)**: Clear motivation and explanation of the computational approaches used, including the selection and implementation of proper tools and methods.
- **Quality of Hands-on Activity (25%)**: Design of a clear and engaging practical activity (well-prepared, easy to follow, illustrating the research question and computational approach).
- **Clarity of Presentation and Organization of Deliverables (20%)**: Clarity and logical flow of the presentation and structure of the code/data and experiments.
- **Professionalism (10%)**: Constructive interaction with colleagues during the workshop and respectful engagment with the work of others.